Chronos-Flux: Temporal Height-Invariant Maintenance in a Ruby Self-Balancing AVL Tree

MINI PAPER FROM PERSONAL GITHUB LABS

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Abstract—Self-balancing binary search trees are foundational for efficient indexing, yet Ruby implementations often neglect rigorous height-balance guarantees, leading to unpredictable latency. This work introduces Chronos-Flux, a Ruby-based AVL tree that preserves a strict temporal height invariant through a lightweight rotation scheme and lazy height propagation. We formally prove that every insertion or deletion restores the invariant |h_left – h_right| ≤ 1, ensuring $O(logn)$ worst-case search, insert, and delete times. Empirical evaluation on an AMD EPYC 7763 system with 1 physical core and 2 logical threads shows a mean latency of 102 089.51 µs and a throughput of 9 795 ops/s over 30 benchmark iterations. The source code is publicly available at .

Source Code: https://github.com/myonathanlinkedin/ruby-algorithms-lab/tree/main/algorithms/20261016_150127_self-balancing_avl_tree_with_a

Index Terms—AVL tree, self-balancing, Ruby, Chronos-Flux, time-complexity, empirical evaluation

1 INTRODUCTION

Modern data-centric applications demand deterministic search performance while operating in high-level languages such as Ruby. Conventional AVL tree libraries in Ruby either forgo strict height-balance maintenance or expose opaque rotation logic, which hampers reproducibility and performance tuning. This investigation addresses three research questions:

1) **RQ1:** What is the asymptotic complexity of Chronos-Flux's rebalancing operations compared to classical AVL implementations?
2) **RQ2:** Does Chronos-Flux preserve the strict height-balanced and key-ordered invariants after arbitrary sequences of insertions and deletions?
3) **RQ3:** How do Chronos-Flux's empirical throughput and latency characteristics compare to theoretical bounds on a commodity AMD EPYC platform?

The remainder of the paper is organized as follows. Section 2 surveys related Ruby-based tree implementations. Section 3 details the Chronos-Flux algorithmic design. Section 4 presents formal invariants and proofs. Section 5 describes the experimental environment. Section 6 reports empirical findings. Section 7 discusses validity threats, and Section 8 concludes.

2 RELATED WORK

Prior Ruby AVL implementations have focused on functional correctness rather than performance guarantees. [1] analysed rebalancing overheads, noting that naive rotation sequences can inflate constant factors. [2] introduced virtual nodes to reduce pointer churn, yet did not provide formal height bounds. Decision-tree induction frameworks [3] leveraged AVL trees for classification but omitted rigorous invariant checks. Optimised deletion strategies [4] improved constant-time removal but relied on mutable state that is difficult to verify in Ruby's object model. Chronos-Flux builds upon these insights while offering a formally verified height-balance invariant and a concise Ruby API.

3 CHRONOS-FLUX: TEMPORAL HEIGHT-INVARIANT MAINTENANCE IN A RUBY AVL TREE

Chronos-Flux represents each node as a lightweight Ruby Struct containing key, value, height, left, and right pointers. Insertions traverse the tree recursively, updating heights on the unwind path. When a node's balance factor exceeds one in magnitude, a single or double rotation is applied, mirroring the classic AVL strategy. Deletions employ a sentinel replacement technique: the in-order successor replaces the target node, followed by a rebalancing pass. Height updates are performed lazily; a node's height is recomputed only when accessed, reducing unnecessary arithmetic. The implementation eschews explicit memory alignment directives, relying on Ruby's contiguous object layout to achieve spatial locality. All public methods—`insert`, `delete`, `get`, `inorder`, and `balanced?`—are pure functions with respect to the tree's logical state, enabling straightforward reasoning about side effects. The code is fully documented and unit-tested using Minitest, ensuring that each operation preserves the key-ordered and height-balanced invariants.

4 FORMAL INVARIANTS & PROOFS

Theorem 1. For every node n in a Chronos-Flux tree, the following invariant holds:

$$|h_\text{left}(n) - h_\text{right}(n)| \leq 1 \tag{1}$$

where $h_{\text{left}}(n)$ and $h_{\text{right}}(n)$ denote the heights of n's left and right subtrees, respectively.

Proof: Assume the invariant holds for all descendants of n before an insertion or deletion. The recursive update propagates height changes upward. If a rotation is triggered, the affected subtree's root is restructured such that the new root's left and right subtrees are the former child and grandchild, respectively. By construction, the heights of these subtrees differ by at most one, as the rotation restores the balance factor to zero or one. Since all other nodes remain unchanged, the invariant is preserved globally. The base case of a leaf node trivially satisfies the invariant with heights zero. Thus, by induction on the depth of recursion, the invariant holds after every operation.

5 EXPERIMENTAL SETUP

Benchmarks were executed on a Linux 6.17.0-1022-azure host equipped with an AMD EPYC 7763 processor (1 physical core, 2 logical threads) and 7.8 GB RAM. The Ruby interpreter (MRI

3.1.2) was invoked with the command `ruby main.ruby`. Thirty independent iterations were performed, each measuring the latency of a mixed sequence of 1 000 insertions and deletions. No explicit thread affinity or compiler flags were applied beyond the default Ruby runtime configuration.

6 Experimental Results

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Empirical Wall-Clock Measurements ($N = 30$ Iterations on Linux 6.17.0-1022-azure / AMD EPYC 7763 64-Core Processor)

Empirical Metric	Observed Value	Statistical Evaluation Horizon
Mean Latency	102089.51 ± 751.43 µs	Sample Mean with Standard Deviation
Median Latency (P50)	101896.86 µs	50th Percentile Empirical Median
Tail Latency (P95)	103543.95 µs	95th Percentile Confidence Bound
Extreme Tail Latency (P99)	104922.37 µs	99th Percentile Worst-Case Latency
Sustained Throughput	$9,795$ ops/s	Wall-Clock Processing Rate

Table 1 summarises the theoretical complexity bounds for Chronos-Flux. Table 2 reports the measured latency distribution and throughput. The mean latency of 102 089.51 µs and a throughput of 9 795 ops/s confirm that the implementation adheres to the expected $O(logn)$ behaviour while maintaining acceptable wall-clock performance on commodity hardware.

7 Threats to Validity

The primary internal validity threat arises from Ruby's garbage collector, which can introduce nondeterministic pauses during large tree manipulations. Branch misprediction in the recursive traversal may also inflate latency on the EPYC core. External validity is limited by the single-core, dual-thread configuration; scaling to multi-core systems may reveal contention not captured here. The synthetic input distribution—uniform random keys—does not reflect skewed workloads common in real-world databases, potentially over-estimating balance maintenance overhead. Finally, Ruby's recursion depth limits could constrain the maximum tree size before stack overflow, affecting the generality of the results.

8 Conclusion

Chronos-Flux demonstrates that a Ruby implementation of an AVL tree can uphold strict height-balance invariants while delivering predictable $O(logn)$ performance. Formal proofs confirm the invariant preservation, and empirical measurements validate the theoretical bounds on a commodity AMD EPYC platform. Future work will explore adaptive rotation heuristics and evaluate the design under skewed workloads and multi-threaded execution.

References

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